

Saketh Velidimalla

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Applied AI systems and data engineer with production ownership of evaluation, observability, analytics, and governed data layers for an enterprise RAG platform serving approximately 140,000 users. Builds measurable quality infrastructure, turns fragmented operational evidence into reliable data products, and translates findings into engineering and business decisions.

Core Competencies

- **Applied AI Systems:** RAG pipelines, LLM integration, retrieval and vector search, FastAPI, Docker, AWS Bedrock, LLM classification
- **Evaluation & Observability:** LLM-as-a-Judge, regression testing, Arize Phoenix, trace analysis, hallucination and retrieval-quality analysis, red teaming
- **Data Products & Engineering:** Python, SQL, PySpark, Airflow DAGs, DynamoDB, Databricks, Delta/Parquet, AWS S3/EMR/Redshift, schema validation
- **Enterprise Analytics & Governance:** Power BI/DAX, semantic models, ServiceNow analytics, metric lineage, PII classification, retention guidance, runbooks

Professional Experience

Johnson & Johnson

Data Scientist

New Brunswick, NJ

Feb 2025–Present

- Owned evaluation, analytics, reporting, and observability for a production enterprise RAG system serving approximately 140,000 internal users across knowledge retrieval, case-management, and content-governance workflows
- Built a three-layer LLM-as-judge pipeline on AWS Bedrock / Llama 3.2 combining rule-based classifiers, batch LLM categorization, and keyword fallbacks to evaluate thousands of queries daily
- Built LLM regression tests with ground-truth benchmarks, judge-based similarity scoring, and cross-run tracking to detect relevance, hallucination, latency, and retrieval-quality regressions before deployment
- Designed containment, query-resolution, repeat-user, and satisfaction metrics from session and query evidence, then used cross-system analysis to separate immediate friction from delayed unresolved needs.
- Connected enterprise RAG traces, DynamoDB records, API payloads, and ServiceNow workflow tables to separate AI, routing, knowledge, and downstream-intake failure modes with row-level evidence
- Built DynamoDB-to-Databricks data pipelines and Power BI semantic models with schema checks, Delta/Parquet validation, KPI definitions, lineage notes, and leadership reporting
- Integrated enterprise AI interaction evidence with unified ServiceNow intake-event data across case creation, external forms, and live-agent channels, preserving row-level Q&A, article, taxonomy, action, target, and case evidence for review
- Conducted cross-system spike analysis linking RAG failures to downstream case patterns, including greeting-drop analysis, no-answer clusters, and same-day versus delayed escalation buckets

Hewlett Packard Inc.

Data Science Engineer

Spring, TX

Jul 2024–Nov 2024

- Built Airflow DAGs running PySpark transformations from S3 PC telemetry into Redshift views for Power BI reporting across hardware, software, power, sensor, and BIOS data
- Optimized PySpark processing on AWS EMR, reducing runtime by approximately 25% through query optimization, repartitioning, and pipeline tuning
- Built engineering KPI reports that helped hardware, software, and chip-design stakeholders compare telemetry trends, prioritize defects, and support roadmap and funding decisions
- Translated raw hardware telemetry into stable analytical tables and Power BI measures so engineering teams could inspect component-level trends without relying on ad-hoc extracts

ECrent Worldwide Company

Machine Learning Engineer

Remote

Jul 2021–Jun 2022

- Built and deployed a content-based recommendation system for real-estate listings using structured attributes and text descriptions to generate personalized suggestions
- Implemented semantic-similarity ranking with Word2Vec embeddings and evaluated results with Precision-at-K and Recall-at-K
- Integrated top-N retrieval into the application backend for real-time Flask API serving
- Prepared feature extraction and evaluation scripts that made recommendation changes measurable before exposing ranked listings through the application workflow

Selected Systems Work

- **SakethWiki:** Local RAG reliability system with curated ingestion, review-gated amendments, telemetry, trace review, evaluation replay, regression gates, and task-based model routing.
- **roastrank-v2:** LLM evaluation and benchmarking platform with rubric-based scoring, regression testing, six-stage output validation, failure labeling, and audit logging. FastAPI, Docker, Hugging Face.
- **SDLC Intelligence Agent:** Built a LangGraph multi-agent workflow that summarizes CI failures, maps errors to source files, flags risky changes, and produces reviewer guidance.

Education & Development

Arizona State University

MS Information Technology (Data Science)

Phoenix, AZ

Aug 2022–May 2024

Completed: Claude Code in Action, Claude 101, Building with Claude API, Agentic AI specialization (Andrew Ng / DeepLearning.AI)

AI tooling: Codex, Claude Code **In progress:** LangGraph Fundamentals, LLM Engineering